Advanced Automation (2025/26)

1st year MEMec, 1st semester 2nd quarter

Project

Project proposal deadline: The class project totalizes 40% of your final grade. The choice of the topic is up to you so long as it clearly pertains to the course material. To ensure that you are on the right track, you will have to **submit a one paragraph description of your project until the end of week 5**. The project proposal should be submitted in Fenix in PDF format until the **12th of December of 2025, at 23:59**.

The projects can focus on four different types: 1) Data Analysis; 2) Monitoring; 3) Hardware; 4) Databases.

In all of the project types, data analysis methods will be required, but depending on the type you/your group wish to pursue different aspects can be considered. The projects can be theoretical derivations or analyses, applications of machine learning methods to problems you are interested in, or something else (to be discussed with course staff). Below you will find examples of project topics for each type, as well as a list of websites where you can search for datasets to use in your project.

You are encouraged to collaborate on the project. We expect a maximum 10-page write-up about the project, which should clearly and succinctly describe the project goal, the dataset, the methods used, as well as present and discuss the results. Each group should submit only one copy of the write-up and include all the names of the group members. The projects will be **graded on the basis of your understanding of the overall course material (not based on, e.g., how brilliantly your method works)**.

Copying will not be tolerated. If you collaborate, you must indicate all of your collaborators.

Grading: The class project will be graded on a scale from 0 to 20. Individual members of a group may receive different grades. As you make progress on your project, you should regularly upload updates to your group's GitHub repository. The frequency and consistency of your GitHub usage will be evaluated. Students who do not use GitHub regularly may lose up to 1 point out of 20.

For clarity, *regular* use refers to making multiple commits over the course of the project to document ongoing progress; it does not imply daily updates.

Submission: The projects are due in **Week 7, until the 14th of January 2025, at 23:59**. Electronic submission is required, through Fenix system in PDF format. The project content (reports, Jupyter notebooks, coding, data, etc.) should also be delivered in your group's GitHub repository. Please paste in your PDF report the link to the respective GitHub.

Useful resources to find datasets

The following websites aggregate and make available datasets for public use. You can use them to search for datasets related to topics you are interested in.

- **Kaggle**: A community-driven platform offering thousands of datasets across various domains, including government, sports, medicine, and finance.
- **Google Dataset Search**: A specialized search engine that helps users discover datasets stored across the web, providing a comprehensive overview of available data.
- **Data.gov**: The U.S. government's open data portal, offering access to over 200,000 datasets covering topics like climate change, crime, and more.
- **UC Irvine Machine Learning Repository**: A collection of databases, domain theories, and datasets widely used by the machine learning community for empirical research.
- **World Bank Open Data**: Provides free and open access to global development data, including statistics on world development indicators.
- **FiveThirtyEight Data**: Offers datasets and code used to create forecasts and graphics across various domains like sports, elections, economics, and culture.
- **IEEE DataPort**: Managed by IEEE (Institute of Electrical and Electronics Engineers), IEEE DataPort is a comprehensive repository for datasets contributed by researchers, professionals, and academics across diverse fields, including engineering, science, and technology.

Project example – Data Analysis

General description: Develop a forecasting system for weather prediction in Sintra.

Objectives: Build predictive models for temperature or precipitation. Compare a minimum of 4 different methods (one for each group member).

Methods: Use feature selection, data reduction or other related methods. Linear or nonlinear methods, such as Linear regression, or neural networks, or SVMs. Choose adequate error measures.

Project example – Monitoring

General description: Develop a monitoring system for weather forecasting in Lisbon, using IPMA API or other weather data feeder.

Objectives: Build a dashboard to show information regarding weather in Lisbon, in real time. Statistical Analysis of the data presented in the dashboard should be done. Use or build an API for collecting data.

Methods: Dash Plotly or Power BI. Python, Statistical analysis

Project example – Hardware

General description: Develop a forecasting system for weather prediction in Lisbon using sensor data acquired with an Arduino.

Objectives: Build a sensor array using an Arduino to record, save and transmit data. Build a predictive model for temperature and/or precipitation.

Methods: Use feature selection, data reduction or other related methods. Use linear methods, such as Linear regression, and implement the final model in Arduino. Feed the data to the model using temperature sensor with the Arduino and the rest of the necessary data can be acquired using an API.

Project example – Databases

General description: Climate change, driven largely by human-induced greenhouse gas emissions, is reshaping temperature patterns across the planet. One of the clearest indicators of this process is the persistent rise in global surface temperatures, documented through long-term instrumental records and supported by a wide range of observational evidence. Examining temperature data from major cities offers a practical way to study regional manifestations of climate change and to understand how local conditions evolve over time.

In this project, the goal is to build an SQL database using the dataset available at: *https://www.kaggle.com/sudalairajkumar/daily-temperature-of-major-cities*

After constructing the database, analyze temperature data from at least four European capital cities. The objective is to explore how their temperature patterns are related and to identify which cities have shown the greatest variation over the past decades.

Objectives: Build and use a SQL database. Analyze temperature data from different cities in Europe. Compare a minimum of 4 different methods (one for each group member).

Methods: Analyze missing values and outliers. Fill missing temperature values with surrounding cities information or other relevant methods, such as KNN.